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```python
import os
import numpy as np
import pandas as pd
from scipy.signal import savgol_filter
from scipy.special import gamma
from scipy.interpolate import interp1d

# ==================== 可调参数区 ====================
DATA_DIR = "./data/HUST data"          # 改成 HUST 原始数据路径
OUTPUT_DIR = "./denoised_for_matrix/InitialParameters/HUST"    # 输出到 HUST 文件夹
# 必须和 HighDimensionMatrix-PINN_Combination.py 里的列名一致
FEATURE_COLS = [
    "voltage mean", "voltage std", "voltage kurtosis", "voltage skewness",
    "CC Q", "CC charge time", "voltage slope", "voltage entropy",
    "current mean", "current std", "current kurtosis", "current skewness",
    "CV Q", "CV charge time", "current slope", "current entropy",
]
CAPACITY_COL = "capacity"               # 容量列不降噪，原样保留

# ---- Ma et al. 2025 SG-GCL 参数 ----
SG_WINDOW = 11                          # 可调：窗口长度（奇数）
SG_POLY = 2                             # 可调：多项式阶数
GCL_SIGMA = 0.08                        # 可调：GCL 核宽度
GCL_ALPHA = 1.5                         # 可调：GCL 形状参数
GCL_ITER = 3                            # 可调：迭代次数

# ---- Huang et al. 2025 多尺度参数 ----
SCALES = [2, 4, 8, 16]                  # 可调：池化尺度
FUSION_MODE = "weighted"                # 可调："average" 或 "weighted"
SCALE_WEIGHTS = [0.4, 0.3, 0.2, 0.1]    # 可调：各尺度权重
# ====================================================

os.makedirs(OUTPUT_DIR, exist_ok=True)

```

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```python
#!/usr/bin/env python3
"""
True combination of Wang et al. 2024 (PINN) and Zhang et al. 2025 (BatLiNet)
on denoised features. Residual dual-path design so matrix does not break physics.

Usage:
    python true_wang_zhang_combination.py --data-dir ./denoised_for_matrix --epochs 100
"""

from __future__ import annotations
import argparse
import os
import random
import numpy as np
import pandas as pd
import torch
import torch.nn as nn
from torch.autograd import grad
from sklearn.model_selection import train_test_split

# ==================== 可调参数区 ====================
FEATURE_COLUMNS = [
    "voltage mean", "voltage std", "voltage kurtosis", "voltage skewness",
    "CC Q", "CC charge time", "voltage slope", "voltage entropy",
    "current mean", "current std", "current kurtosis", "current skewness",
    "CV Q", "CV charge time", "current slope", "current entropy",
]
NOMINAL_CAPACITY = 1.1          # 可调：标称容量 Ah
N_FEATURES = 16
WINDOW = 16                     # 可调：TimesNet 因果窗口长度
K_PERIODS = 3                   # 可调：保留的 top-k 周期数
MATRIX_H, MATRIX_W = 8, 8       # 可调：矩阵 resize 后的高宽
HIDDEN = 60                     # 可调：Wang 隐藏层宽度
ALPHA_PHYS = 0.1                # 可调：物理损失权重（对应 Wang loss2）
BETA_MONO = 0.01                # 可调：单调性损失权重（对应 Wang loss3）
MIX_INIT = 0.5                  # 可调：Wang/Zhang 初始混合权重
SEED = 420
EPOCHS = 100                    # 可调：训练轮数
BATCH_SIZE = 32                 # 可调：批大小，越大 GPU 利用率越高
# ====================================================
```

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```
The 17-d vector (16 features + cycle index) is min-max scaled to [-1, 1]
inside each battery, as in Wang's loader. The period matrix is built from the
cleaned raw feature values before that scaling.

The split is leave-one-condition, not the PINN rule "battery id % 5 == 0".
MIT conditions are the three date folders. HUST conditions are the number
before the hyphen in the filename. Of the remaining batteries, 20% is
validation (sklearn train_test_split, random_state=420). Test labels are not
used to choose the checkpoint. This is not a controlled comparison with the
PINN baseline or with the published BatLiNet cell lists, and these CSVs are
not the raw Q-V curves used by Zhang et al.
"""

from __future__ import annotations

import argparse
import json
import os
import platform
import random
import subprocess
import time

import numpy as np
import pandas as pd
import torch
import torch.nn as nn
from sklearn.model_selection import train_test_split
from torch.autograd import grad
from torch.nn.functional import interpolate

# ==================== 可调参数区 ====================
DEFAULT_EPOCHS = 100            # 训练轮数
DEFAULT_EARLY_STOP = 70         # 早停
DEFAULT_BATCH_SIZE = 64         # 批大小
DEFAULT_WINDOW = 16             # 窗口长度
DEFAULT_STRIDE = 5              # 步长
DEFAULT_K = 2                   # 周期数
DEFAULT_HEIGHT = 8              # 矩阵高
DEFAULT_WIDTH = 8               # 矩阵宽
DEFAULT_N_REFS = 2              # 参考电池数
DEFAULT_SEED = 420              # 随机种子
# ====================================================
```

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```
#!/usr/bin/env python3
"""
SHAP explanations for the true Wang + Zhang combined model.
Based on Molnar (2019) "Shapley Values" chapter (Model-Agnostic Methods).

Concepts used:
- Efficiency: sum of SHAP values = prediction - average prediction
- Local explanation for single battery-cycle
- Global feature importance by averaging absolute SHAP
- Monte-Carlo style approximation via KernelSHAP

Usage:
    python shap_explanation.py --data-dir ./denoised_for_matrix --model-path best_combined.pt
"""

import argparse
import os
import numpy as np
import pandas as pd
import torch
import shap
from sklearn.model_selection import train_test_split

# ==================== 可调参数区 ====================
FEATURE_COLUMNS = [
    "voltage mean", "voltage std", "voltage kurtosis", "voltage skewness",
    "CC Q", "CC charge time", "voltage slope", "voltage entropy",
    "current mean", "current std", "current kurtosis", "current skewness",
    "CV Q", "CV charge time", "current slope", "current entropy",
]
NOMINAL_CAPACITY = 1.1          # 可调: 标称容量
N_FEATURES = 16
WINDOW = 16                     # 可调: 与结合模型一致
K_PERIODS = 3                   # 可调
MATRIX_H, MATRIX_W = 8, 8       # 可调
HIDDEN = 60                     # 可调: 与结合模型一致
SEED = 420
N_BACKGROUND = 50               # 可调: KernelSHAP 背景样本数（书里 Monte-Carlo 的 M）
N_EXPLAIN = 20                  # 可调: 要解释的测试样本数
# ====================================================

# -------------------- 复用结合模型的关键组件（接口） --------------------
```

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```python
#!/usr/bin/env python3
"""
SHAP explainability for HighDimensionMatrix-PINN_Combination.py (Wang SolutionU path).

Usage:
    python SHAP_HighDimension.py --model-path <best_wang_valid.pt> --dataset MIT
"""

import argparse
import os
import sys
import numpy as np
import torch
import shap

# ==================== 可调参数区 ====================
N_BACKGROUND = 80
N_EXPLAIN = 30
NSAMPLES = 100
WINDOW = 16
K_PERIODS = 2
HEIGHT = 8
WIDTH = 8
# ====================================================

# 确保能导入同目录下的模型文件
```